Junior Roller Skate - Rear Brake Housing

Target product brief

Product context

The company is developing the next generation of a rear brake housing for a junior roller skate used by children aged 6 to 12. The component protects and retains the rear braking mechanism and is exposed to drops, falls, kerbs and outdoor conditions.

Product objectives and priorities

- The new component should provide at least the same level of protection and durability as the current version.
- The main problem to solve is cracking reported during winter use, particularly after impacts or falls.
- The expected outdoor-use range is approximately -10°C to 45°C.
- A mass reduction of approximately 15% would be commercially valuable, but reliability has higher priority.
- The component should contain at least 30% recycled material by mass.
- The visible surface should remain black and low-gloss and should not look noticeably lower quality.
- The team wants to avoid a significant increase in component cost or disruption to manufacturing.
- The local geometry around the screw boss is still being reviewed and may change.

Current component and industrial context

Field	Information	Status
Component mass	82 g	Confirmed
Nominal wall thickness	2.5 mm	Confirmed
Manufacturing process	Injection moulding	Confirmed
Current material	Impact-modified polypropylene; exact grade unknown	Partially known
Current material density	Unknown	Not available
Current material price	Approximately €2.10/kg	Estimate
Declared recycled content	0%	Confirmed
Assembly	Two screws and one snap-fit	Confirmed
Annual volume	Approximately 180,000 parts	Forecast
Current cycle time	38 seconds	Confirmed
Current scrap rate	Approximately 3%	Estimate
Existing mould	Strong preference to reuse it	Stakeholder preference

No standardized material-specimen data is available for the current grade. Existing evidence comes from finished parts, field returns and internal observations.

Field issue and available evidence

Customer returns

- 17 cracked components were returned during the previous winter season.
- 13 of the 17 returns came from regions where temperatures had fallen below 0°C.
- The real impact conditions and product history are unknown.
- Most cracks appeared near the screw boss or at the junction between the main wall and the brake support.

Internal cold drop observation

- Five complete rollers were conditioned for four hours at -10°C.
- Each roller was dropped from 1.0 metre onto concrete in several informal orientations.
- Two of the five brake housings cracked.
- No room-temperature control group was tested.
- The exact impact energy was not measured and the protocol is not standardized.

Assembly observations

- No immediate cracking was observed during assembly.
- Visible whitening sometimes appears around the screw boss.
- Screw torque is not systematically recorded.

Stakeholder notes

Product manager

Reliability is the first priority. The component must not feel less robust than the current one. A lighter part is valuable, and the visible surface must remain black and low-gloss.

Product engineer

The root cause may involve the material, local geometry, assembly stress or a combination. The component must not deform enough to interfere with the brake mechanism. No numerical stiffness target exists.

Manufacturing engineer

The current mould and machine were developed for an injection-moulded polypropylene. Large changes in shrinkage may create dimensional or assembly issues. Abrasive reinforcements may increase mould wear. A cycle-time increase above 10% should be avoided. The mould has thin flow sections, but no validated minimum flow requirement is available.

Sustainability manager

At least 30% recycled content by mass is expected. Post-consumer recycled content is preferred. Feedstock origin and traceability must be documented. Recycled content is not a success if durability decreases significantly.

Supplier suggestion

> *“We recommend a 30% glass-fibre-reinforced PA6 because it is stronger and should solve the issue. We have not tested it in the current mould and have not provided supporting data for this application.”*

This statement is part of the product dossier. Treat it according to the level of evidence you consider appropriate.